



Zeina BOURHANE
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I. Academic position

[11/2024 – Present] Postdoctoral fellow

Identification of Bioactive molecules in Thermal Spring Waters for skin microbiome enhancement

CARUSO Project: Interreg POCTEFA

Institut des sciences analytiques et de physico-chimie pour l'environnement et les matériaux
Pau University (UPPA), **CNRS, France**

Advisors: Prof. Robert Duran

[2023] Qualification- Section 68: Assistant Professor by the Ministry of Higher Education and Research, **France**

[06/2023 – 07/2024] Researcher

Microbial Ecology (CNRS-L - UNESCO)

Laboratoire de Microbiologie, Santé et Environnement

Lebanese University, Lebanon

[02/2022 – 05/2024] Postdoctoral fellow

Modelling of chemical and microbiological processes in the atmosphere and soil

MOBIDIC Project: Make Our Planet Great Again (MOPGA)

Clermont-Ferrand Institute of Chemistry (ICCF)

Clermont Auvergne University (UCA), **CNRS, France**

Advisor: Dr. Barbara Ervens

[09/2018 – 12/2021] Doctoral student

Thesis: Anthropogenic impacts on microbial communities in agriculture arid zones

AQUASALT Project: ERANETMED Program

Institute of Experimental Biocenotics and Agrosystems (IBEAS)

Pau University (UPPA), **CNRS, France**

Advisors: Prof. Robert Duran, Dr. Christine Cagnon

II. Education

[09/2018 – 01/2022] Ph.D. in Physiology and Biology of organisms-Populations-Interactions

Pau University (UPPA), **France**

[09/2016 – 06/2018] MSc. in Management and Conservation of Natural Resources

Lebanese University, Beirut, **Lebanon**

III. International Scientific stays

[09/2026 – 11/2026] Research training (Confirmed: FEMS)

Metagenomic insights into microbial dynamics under redox conditions in extreme environments

Host: Pr. Carmen Castañeda, **CSIC, Zaragoza, Spain**

[09/2024] International School on Ion Mobility MASS SPECTROMETRY

Ion Mobility mass spectrometry: Fundamentals, Advances, and Applications

Host: Ettore Majorna Foundation, **Erice, Italy**

[05/2022] UNESCO-MAB Laureate

[10/2019 – 02/2020] Doctoral Mobility

Adapt microgAMBI microbial Index for contaminated soils and sediments

AZTI Marine research center, **Pasaia, Spain**

Hosts: Prof. Angel Borga, Dr. Anders Lanzén

IV. Peer reviewed publications

Bourhane Z., Lanzén A., Cagnon C., Ben Said O., Mahmoudi E., Coulon F., Atai E., Borja A., Cravo-Laureau C., Duran R. (2022). **Microbial diversity alteration reveals biomarkers of contamination in soil-river-lake continuum**, *Journal of Hazardous Materials*.

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. et al. (2023). **Vertical organisation of microbial communities in hypersaline wetland, Salineta**, *Frontiers in Microbiology*.

Amato P., Mathonat F., Nunez L., Péguilhan R., **Bourhane Z.**, Rossi F., Vyskocil J., Joly M., Ervens B. (2023). **The aeromicrobiome: the selective and dynamic outer-layer of the Earth's microbiome**, *Frontiers in Microbiology*.

Bourhane Z., Amato P., Ervens B. (2025). **Comparison of polycyclic aromatic hydrocarbons (PAHs) biodegradation in the atmosphere and soil**.

Liu Bang., **Bourhane Z.**, Yoa Jun P., Castañeda C., Cagnon C., Cravo-Laureau C., Chen Z., Duran R. (2026). **Genome-resolved analyses disentangle the strategies of hypersaline microbiomes and metabolic interactions within a hydrological gradient**. (Under review: Submitted to Environmental Science & Technology)

Aziza Kouchou, **Bourhane Z.**, Fadwa Tadist, Olfa Ben Said, Naoual Rais, Robert Duran, Naïma El Ghachtouli (2026). **Effects of pollution on the activity of the soil/sediment microbial communities from a soil-river-lake continuum in Ichkeul Lagoon**. (Under review)

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. (2026). **Metagenomic insights into microbial dynamics under redox conditions in a hypersaline wetland soil using Indicator of Redox in Soils (IRIS) films**. (Under review)

Bourhane Z., Duran R., Cravo-Laureau C. (2026). **Unveiling the metagenomic profiles of Pyrenean thermal waters**. (Under review)

V. Oral presentations

Bourhane Z., Cravo-Laureau C., Meselier M., Pinaly H., Duran R., Preud'homme H. *Are bioactive molecules in thermal waters revolutionizing cosmetic products?*. **AFEM: 12th Francophone Association of Microbial Ecology Congress**, Beaune, France, 18-21 November **2025**.

Bourhane Z., Amato P., Ervens B. *A comparison of the loss of polycyclic hydrocarbons (PAHs) in soil and in the atmosphere*. **European Geosciences Union (EGU)**, Vienna, Austria, 23–28 April **2023**.

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. *Microbial diversity alteration reveals biomarkers of contamination in soil-river-lake continuum*. **International Conference: Science and Research in, for and with UNESCO Biosphere Reserves**, Eberswalde, Germany, 16-20 May **2022**.

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. *Role of microbial diversity alterations in revealing biomarkers*

of contamination in soil-river-lake continuum. **EcotoxicoMicYR**, December 2021.

Bourhane Z., Lanzén A., Cagnon C., Ben Said O., Mahmoudi E., Coulon F., Borja A., Cravo-Laureau C., Duran R. *Bacterial bioindicators for reporting contamination in a coastal soil/river/lake continuum*. **SETAC Europe 31st Annual Meeting**, Seville, Spain, 3-6 May 2021.

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. *Role of bioindicators in the environmental assessment*. AZTI, Marine Research, Basque Research and Technology Alliance (BRTA). Pasaia - Gipuzkoa, Spain, Nov. 2019.

VI. Poster presentations

Bourhane Z., Cravo-Laureau C., Meselier M., Pinaly H., Duran R., Preud'homme H. *Novel Bioactive molecules in Thermal Waters: A New Frontier in Cosmetic Innovation*. **EcotoxicoMicYR**, October-November 2025.

Bourhane Z., Cravo-Laureau C., Meselier M., Pinaly H., Duran R., Preud'homme H. *Bioactive molecules in thermal waters: A revolution in cosmetic products*. **Microbes: 20th Congres French Society of Microbiology** Bordeaux, France, 24-26 September 2025.

Bourhane Z., Chevillon V. *Bioactive molecules identification in POCTEFA area plant extracts and thermal spring waters for skin enhancement microbiome in biosourced cosmetic*. **International School on Mass Spectrometry**, Erice, Italy, 9-14 September 2024.

Bourhane Z., Cagnon C., Castañeda C., Rodríguez-Ochoa R., Álvaro-Fuentes J., Cravo-Laureau C., Duran R. *How microbial communities organize vertically in hypersaline wetland to ensure biogeochemical cycles?*. **EcotoxicoMicYR**, December 2023.

Bourhane Z., Ervens B. *Comparison of bacterial biodegradation of polycyclic aromatic hydrocarbons (PAHs) in the atmosphere and soil*. **International Society for Microbial Ecology (ISME18)**, Lausanne, Swiss, 14-19 August 2022.

Bourhane Z., Cagnon C., Castañeda C., Rodríguez-Ochoa R., Álvaro-Fuentes J., Cravo-Laureau C., Duran R. *Vertical organisation of microbial communities in Salineta hypersaline wetland, Spain*. **FEMS Conference on Microbiology**, Belgrade, Serbia, 30-2 July 2022.

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. *Bacterial bioindicators for reporting contamination around Ichkeul Lake*. **EcotoxicoMic 2020**, Second International Conference on Microbial Ecotoxicology, Montpellier, France, 6-9 October 2020.

Research grants and Awards

2026: Research grant. FEMS (Federation of European Microbiological Societies)

2023: Travel grant. AFEM Young researcher mobility

2022: Travel grant. FEMS Conference attendance Grant, Belgrade

2021: Award. Laureate of the **UNESCO MAB for young scientists**

2019: Travel research grant. Grant of Excellence, Energy and Environment Solutions (E₂S), France

2019: Travel research grant. Aquitaine Mobility (AquiMob) International Mobility Grant, UPPA

Membership of scientific organizations

2025- SFM member (Société Française de Microbiologie) : <https://www.sfm-microbiologie.org/>

2025- NextGen4MED: Network of young researchers focused on sustainable water management in the Mediterranean

2024- SFBI member (Société savante de la bioinformatique) : <https://www.sfbi.fr>

2020- EcotoxicoMic member: <https://ecotoxicomic.org>

2019- AFEM member (Francophone Association of Microbial Ecology)

Peer reviewer

Frontiers in microbiology, Environnemental Pollution, Microbial Ecology, Environmental Research, Scientific Reports, Thalassas (Springer)

Technical skills

Bacteria culture

DNA extraction and PCR

Agarose gel electrophoresis

Statistics and bioinformatics: QIIME2, R Studio

Modelling: quantitative prediction

Mass Spectrometry